**Report# : 1**
**Team : Jarvis**
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# I. Introduction

This report outlines the progress of our team for Phase1 of the Bosch Future Mobility Challenge. The **primary focus** during this phase was on setting up the vehicle, understanding the provided software framework, and **achieving stable manual control** using the demo code.

We have initiated autonomous functionality through **stop sign detection**, which is capable of stopping the vehicle, though accuracy improvements are ongoing. In parallel, the simulator environment was successfully set up to visualize the track and vehicle connectivity within the simulation environment. We also **printed out a full scale physical track** encompassing main features of the Competition map. This report summarizes the work completed so far and defines the upcoming tasks toward full autonomy.

# II. Planned Activities (Nov25-Dec22, 2025)

The following activities were planned at the start of the Phase 1 period:

A. Architecture (Ongoing)
B. Project Plan (Completed)
C. Manual Vehicle Control (Completed)
D. Simulator Setup (Completed)
E. Stop Sign Detection (Ongoing)
F. Lane Detection (Ongoing)

### A) Architecture (Ongoing)

A block diagram architecture of the communication between various planned software modules and hardware components was provided by Bosch & was studied. We found it to be quite comprehensive & since we are just starting out, we decided to adapt it **as it is for now**. We plan to finalize any modifications/changes by Phase2.

### B) Project Plan (Completed)

**A project plan** aligned with the above defined architecture and competition timeline **was mapped out,** focusing on a step-by-step progress from manual operation to the various autonomous features involving lane detection & object detection & extra vehicular communication. Adapting the plan to initial setup challenges required minor timeline adjustments.

### C) Manual Vehicle Control (Completed)

The vehicle was successfully operated in manual mode using the provided demo code. **KL mode, Steering and speed control** through the dashboard were verified. **Initial calibration** and configuration required repeated testing.

**D) Simulator Setup (Completed)**

The **Gazebo simulator environment** was successfully configured through **VMware workstation** virtual machine. The track map is visible, and the vehicle is connected correctly. Autonomous task implementation in the simulator has not yet started. Simulator configuration and synchronization with the framework required careful setup.

**E) Stop Sign Detection (Completed)**

A **stop sign detection module was implemented** and integrated with the vehicle control pipeline. **Upon detection, a braking command is sent** through the BFMC **serial interface to the STM32** Nucleo, causing the vehicle to stop physically.Detection accuracy is currently limited under certain lighting conditions. Improving robustness and consistency in real-time operation will be a challenge we anticipate.

**F) Lane Detection (Ongoing)**

A lane detection module using camera input from the car was started using **Canny edge detection** techniques. We should have it working by the next update.

## III. General Status of the Project

The project has progressed steadily with **successful setup** of the physical vehicle, & the **physical track** , a mapping out of the architecture block diagram, project plan roadmap, simulator readiness, and stable manual control. Autonomous functionality has been implemented through stop sign detection integrated with real vehicle braking. The vehicle is capable of detecting a stop sign and performing the required stopping maneuver.

Road sign & Traffic light detection, lane detection, & if time permits intersection detection are planned for upcoming phases to achieve full autonomous operation.

## IV. Upcoming Activities

1. Improve accuracy and reliability of stop sign & other road sign detection.
2. Continue development of traffic light detection.
3. Start lane detection and tracking for straight and curved paths.
4. Intersection detection & navigation